

Milestone Three

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Problem Statement/Description

This paper develops a unified framework for financial crime detection utilizing user and transaction-based features. We apply this framework to the IBM Transactions for Anti-Money Laundering (AML), Paycheck Protection Program (PPP), and the European Credit Card (ECC) Fraud Detection datasets. In the IBM dataset we treat account holders as users and payments as transactions. Likewise, in the PPP dataset we treat businesses as users and loan applications as transactions. Finally, in the European Credit Card Fraud Detection dataset the anonymized features correspond to users, and the amount and time features correspond to transaction features. By analyzing financial crime datasets across three industries, I will identify common issues such as class imbalance, non-linear relationships, and high false positive rates, as well as their solutions. Data scientists can utilize these strategies to combat financial crime in organizations to save money, ensure regulatory compliance, and maintain customer confidence.

Exploratory Data Analysis

Preprocessing

The IBM dataset comprises 18 files. For this paper, I utilized the small-high dataset, which contains over 500,000 instances and 5,000 instances of fraud with account and transaction features files. I did not merge the files since the account features only contain high cardinality categorical data. The transaction file contains 18 duplicate transactions, but I retained these since duplicate transactions could correlate with fraudulent activity. Additionally, there are no missing values. I utilized one-hot encoding to convert three categorical features, Payment Currency, Receiving Currency, and Payment Format, into numeric formats. For the Amount Received/Paid features there is a large range of values, from 0.00 to over 36 trillion. The amounts represent 15 different currencies, so currencies like Yen or Rupee have much larger scales than the US Dollar.

I standardized the currencies to USD by utilizing historical conversion rate data. I dropped the original amounts from the dataset and engineered a categorical feature representing the original currency. In the boxplots of each feature, we see outliers for every feature. The largest outliers are for the currency standardized Amount Received/Paid features. I retained the outliers since extremely high values could correlate with fraudulent transactions. Finally, the numeric features must be standardized prior to modeling. For the Amount Received/Paid features, I used interquartile standardization, which is more robust to outliers and standard scaling for the others.

The ECC dataset does not contain any missing values and the majority of the features are already standardized. The author transformed these features using principal components analysis to protect client confidentiality. The only non-transformed features are Time, Amount, and Class. The label, Time, and Amount are already in a numeric format, so no type conversion or encoding is required. The dataset contains 1,854 duplicate rows, but I retained them since duplicate transactions could indicate fraudulent activity and 32 of these rows represented the positive class (over 6% of total positive instances). The Amount feature has many outliers as the majority of transactions are for small amounts. I retained the extreme values since the scale of the outliers makes sense in context of credit card transactions. Additionally, the majority of the PCA transformed features had outliers. Prior to modeling, I recommend interquartile scaling for the Amount feature due to the scale of outliers and standard scaling for the other features.

The PPP loan dataset does not contain missing values or duplicate instances. 13 geographic-based and demographic-based status categorical features can be encoded using one-hot or ordinal encoding. I utilized ordinal encoding since the dataset contains over 1 million rows and many categorical features were high cardinality, which would result in a highly dimensional dataset. Three date features, DateApproved, ForgivenessDate, and LoanStatusDate,

required encoding from dates to numerics in order to be used for modeling. The dataset is split into two files, one containing 100 positive cases of fraudulent loans with the corresponding business name and the other containing over 900,000 PPP loan applications. I attempted to join these datasets by business name. When validating the quality of the join, I found only 83 out of the 100 fraud cases were found in the joined dataset, resulting in a smaller number of positive instances to analyze and train models on. There were four rows in the dataset with infinite numeric values. I replaced these with null values instead. Next, InitialApprovalAmount, CurrentApprovalAmount, NotForgivenAmount, and a number of the proceed features had a common maximum value of 10,000,000.00, which is a business rule due to PPP loan maximum being \$10,000,000 for an individual borrower. Finally, for CurrentApprovalAmount, the \$150,000 minimum value looks to be clipped. It likely indicates all loans \$150,000 or lower.

Univariate Analysis

In the IBM histograms for the Amount Received/Paid features the distribution is heavily skewed with a large range of values. The minimum is \$0.00 USD, with a 75th percentile of \$5,084.63 and a maximum of over \$34.6 billion. To address the skew a log transformation can be utilized, and to ensure the outliers do not affect scaling we can use strategies such as interquartile scaling. Next, in a bar chart of Payment Currency and Payment Format, the majority of transactions used US Dollars or Euro and Cheque or Credit Card (see Figure 1). Since the payment currencies and formats are unbalanced, models performance may be impacted as models could generalize poorly to minority categories. Finally, all the numeric features should be standardized to ensure different scaled features do not affect model performance.

In the ECC dataset histograms, the PCA transformed features are normally distributed and were standardized prior to transformation. The Amount feature is right skewed, with a

minimum of 0.00, 75th percentile of 77.17, and maximum of 25,691.00. To address the skew, a log transformation can be applied. The Time feature is bimodal, with peaks around 60,000 and 150,000, suggesting that there are two periods for increased transactions. These features require standardization due to their differing scales. Finally, the Class feature is highly unbalanced, with over 99.998% of transactions being non-fraudulent. Recall or f1 must be a key evaluation metric in model selection. Models with high precision and low recall will consistently predict non-fraudulent due to this class imbalance. Additionally, oversampling can help models to learn the positive class better by increasing the ratio of positive to negative instances.

The PPP dataset contains many right skewed features such as JobsReported and the majority of amount and proceed features. Additionally, the majority of the percentage features like ForgivenPercentage are left skewed. A log transformation on the skewed features should be done prior to standardization and modeling. In the histogram of the target variable there is a large class imbalance. Only 83 of 968,000+ instances are in the positive class. Oversampling and evaluation metrics like recall can address the class imbalance. In bar charts of the BorrowerState and BusinessAgeDescription, applications were more popular in higher populated states such as California and for businesses that are greater than two years old. The most common types of businesses are Corporations and LLC. Finally, Ohio and California serviced the most PPP loans. Models could generalize poorly to the minority classes in these categorical features. Summary statistics show that features do not have the same scale. It will be important to standardize the features prior to modeling to ensure higher scaled features do not overpower others.

Bivariate Analysis

The IBM dataset correlation matrix shows that Payment and Receiving Currency are highly positively correlated. Only 0.014% of instances have differing Receiving Currency and

Payment Currency. Additionally, all cases of money laundering in the dataset are for payments with matching Payment and Receiving Currency, so this correlation could be useful in models. As shown in Figure 2, different payment amounts are more common with certain payment formats. For example, in the bin in the histogram for transactions close to \$0, Reinvestment and Credit Card are the most popular; whereas for very large transactions, ACH is the most common. In a bar plot of the count of fraudulent transactions grouped by the payment format shown in Figure 3, some formats are more likely to be fraudulent than others. ACH is the majority payment format for fraudulent transactions, even though ACH is not the majority category in the dataset. Calculating the correlation between the ACH payment format and the label, the correlation is 0.753 in a balanced sample of 50% fraudulent to non-fraudulent transactions. This verifies that the payment format is extremely important for detecting fraudulent transactions in this dataset.

The ECC dataset correlation matrix shows six strong correlations with the target. Initially, by creating a correlation matrix of the entire dataset we see that the target variable has strong negative correlations with V12, V14 and V17. Next, I took a balanced sample of 50% fraudulent and 50% non-fraudulent transactions. As shown in Figure 4 there are additional positive correlations between V2, V4, and V11 and the target. Additionally, we can see that features such as V23 and V25 do not have correlation with the target. These features could likely be ignored during model selection. As shown in Figure 6 there is a clear separation of the target using a linear relationship between the strongly correlated features. This gives a strong indication that these will be important features in models. In boxplots of V4 and V11 for each of the target values, we see in V11 that there are more outliers for fraudulent transactions than non-fraudulent (see Figure 7). I recommend removing these outliers, as they could affect the performance.

The correlation matrix of the entire PPP dataset does not show any correlations with the target, however a balanced sample of 50% fraudulent instances shows 7 strong correlations (see Figure 5). The strong negative correlations are between the target and ProcessingMethod, forgiveness features, and loan status features and there is a strong positive correlation with NotForgivenAmount. When looking at the boxplot of ForgivenessDate grouped by the target variable, we see that the majority of non-fraudulent instances have a ForgivenessDate corresponding to the range of 2022 and fraudulent instances correspond to 1900, a placeholder value for non-forgiven (see Figure 8). This indicates fraudulent loans were generally not forgiven. These two correlations seem to leak information about the target variable. Similarly, LoanStatus and LoanStatusDate indicate when a loan is dispersed, leaking similar information about rejected loans. I recommend dropping these features prior to building models. As a result, the strongest correlations with the target are all likely to be unusable and non-linear relationships will need to be utilized in modeling to make use of less strong correlations.

Data Conclusions

Expected/Unexpected Data

Through analysis of the transaction features in the IBM dataset, we saw that Payment Amount, Payment Format, Receiving Amount, and Receiving Format are likely to be important features in models to predict money laundering. These features seem like expected candidates since prices and payment formats are the main variations in transactions. Additionally, we found that the largest preprocessing effort was for currency conversions to standardize the scale of transactions, as well as one-hot encoding for categorical features. Finally, it was unexpected to see that the correlation matrix does not differ greatly when plotting the correlation between an oversample of the positive class versus the entire dataset.

Through the analysis of the credit card dataset, I identified features V2, V4, V11, V12, V14, and V17 to be strongly correlated with the target. One of the most unexpected things I found in the dataset was the spread of values for features correlated with the target was larger for fraudulent transactions than non-fraudulent. This indicates that many of the fraudulent transactions show anomalous behavior compared to the baseline. Additionally, I saw that in the correlations we could not identify the positive correlations until oversampling was utilized. This shows that the ratio of positive class instances relative to the negative instances needs to be larger to identify the patterns in the dataset. Similar sampling techniques may be useful for training models.

In the PPP dataset the most unexpected data issue was the dataset joining issue to create the target variable. Due to only 83 of the 100 fraud cases being available in the dataset, further analysis on the target was limited. As a result, there could be correlations or other patterns which would have been better identified otherwise. Additionally, the clipping of the CurrentApprovalAmount at \$150,000 was unexpected. Only 3 of the 83 fraudulent instances have this clipped value, so it was not too problematic for the analysis.

Disqualified Data

In the IBM dataset, we eliminated some user level features from potential features for modeling. Bank Name and Entity Name cannot be encoded since they are high-cardinality categorical features, with different values for each row in the dataset. Other user level features like Bank ID, Entity ID, and Account ID could still be useful for modeling with feature engineering, such as representing whether users have previously engaged in transactions, but could not be utilized otherwise.

In the credit card dataset, all of the features were numeric and could be candidates for modeling. However, when analyzing the correlations between each feature and the target, we found that features such as V23 and V25 have no correlation with the target. As a result, when performing feature selection we can verify whether these features are removed from models.

In the PPP loan dataset high cardinality categorical features like LoanNumber, SBAOfficeCode, ServicingLenderLocationID, BorrowerAddress, BorrowerCity, BorrowerZip, BorrowerName, and OriginatingLender can be disqualified since they are not useful for modeling. Next, protected attribute features including Race and Gender should be disqualified since they could result in biased models. Additionally, the most correlated features with the target, ForgivenDate, ForgivenPercentage, LoanStatus, and LoanStatusDate must be disqualified from modeling due to information leakage of the target. Finally, SBAGuarantyPercentage is uniformly distributed with a mean of 100 and standard deviation of 0. It will not be a useful feature as a result and can be excluded.

Supervised/Unsupervised Analysis

The IBM dataset relationships between the label and the other features are non-linear. The dataset includes many different types of money laundering strategies. I recommend first utilizing a baseline interpretable classification model such as Decision Tree to find preliminary results. Next, a more complex ensemble model such as Gradient Boosting Trees and comparison of performance against the baseline model can be performed. The nature of Gradient Boosting Trees, where subsequent trees attempt to predict the error in the previous tree, could help to learn the more complex patterns in this dataset. Finally, it will be important in this dataset to use metrics such as recall or f1 to measure the model performance due to the class imbalance. High precision models can always predict the negative class but are not helpful for predicting fraud.

The analysis of the ECC dataset revealed similar patterns to the IBM dataset. The target requires a classification model. Additionally, recall or f1 should be utilized for model selection criteria to address class imbalance. Finally, ensemble models will be the most useful to uncover the different fraud strategies. In this dataset, model interpretability is a challenge since the many features are a result of PCA transformation, limiting understanding of the original features.

Through the PPP dataset analysis, we found that the target variable is categorical and requires a classification model for supervised learning. The largest challenge with this dataset is the class imbalance, as well as a small number of the positive class instances. There is a risk of overfitting a model to the small number of instances. Leave-one-out cross-validation, oversampling, and a model like Bagging Decision Tree should be utilized. Leave-one-out cross-validation will allow us to use 82 instances for training and 1 for testing n times, to evaluate how the model generalizes, while still using as many data points as possible for training due to the positive class size. Oversampling will allow us to leverage all of the positive classes in our modeling and ensure that models do not become biased towards the majority class which is over 99% of the dataset. Finally, Bagging Classifier will be a good candidate model due to its characteristics to reduce variance and minimize overfitting.

Figure 1: IBM Payment Currency and Format Frequencies

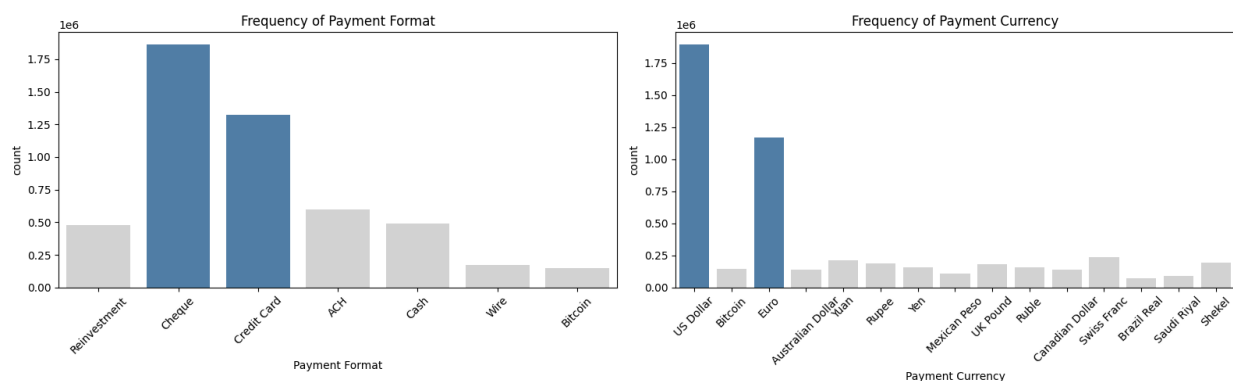


Figure 2: IBM Payment Format by Amount Paid

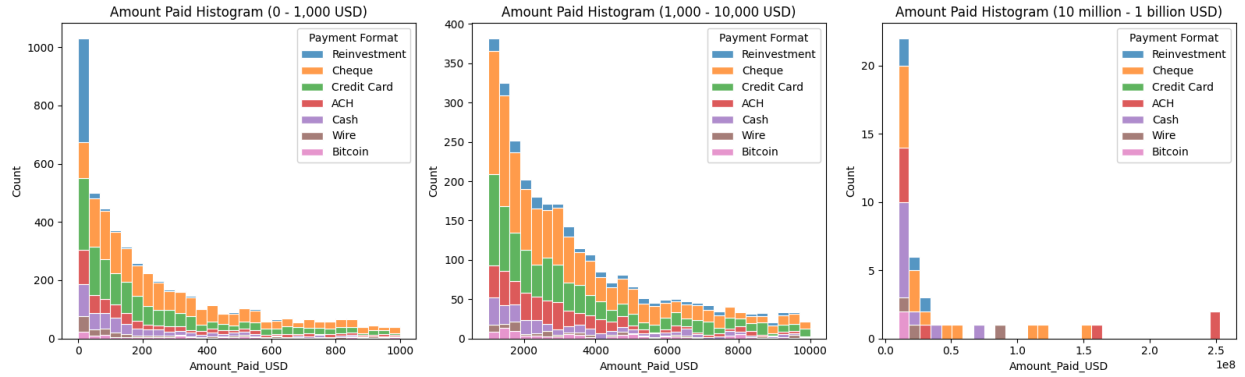


Figure 3: IBM Frequency of Payment Formats for by Target

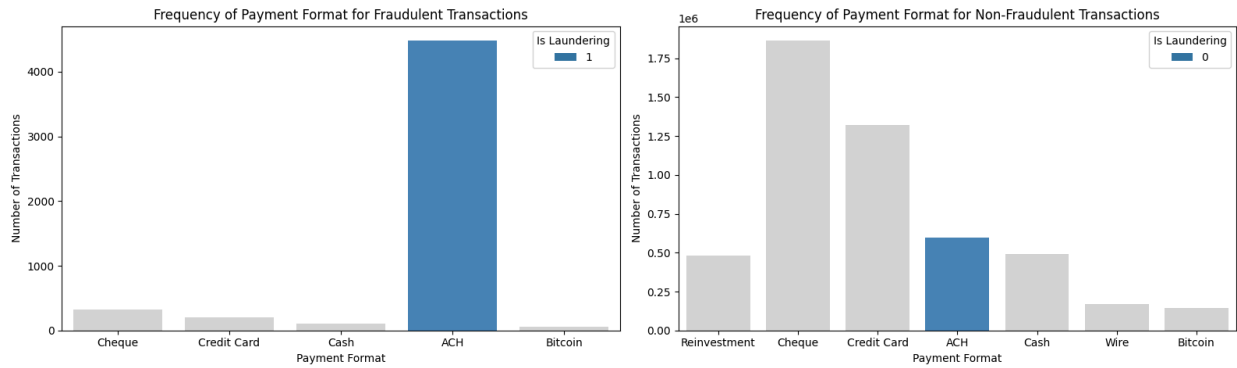


Figure 4: ECC Correlations

Figure 5: PPP Correlations

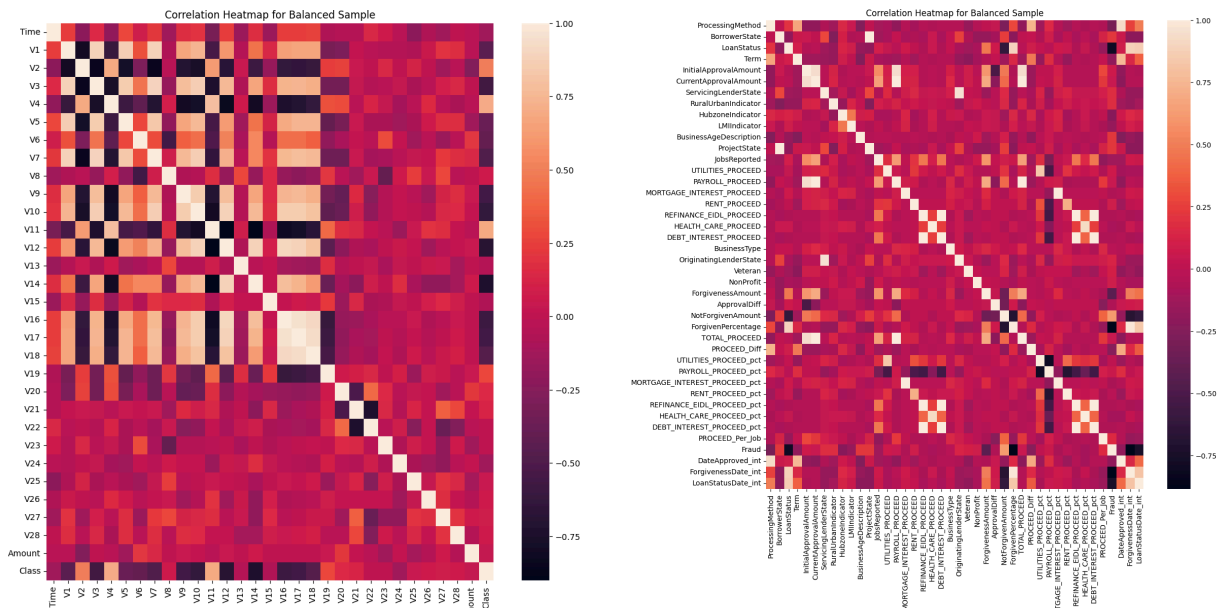


Figure 6: ECC Scatterplot Colored by Target

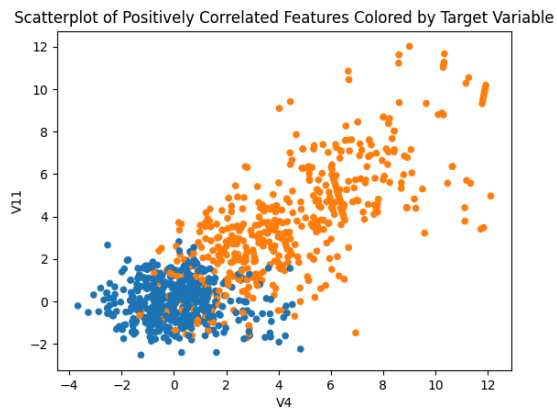


Figure 7: ECC Boxplot of Features by Target

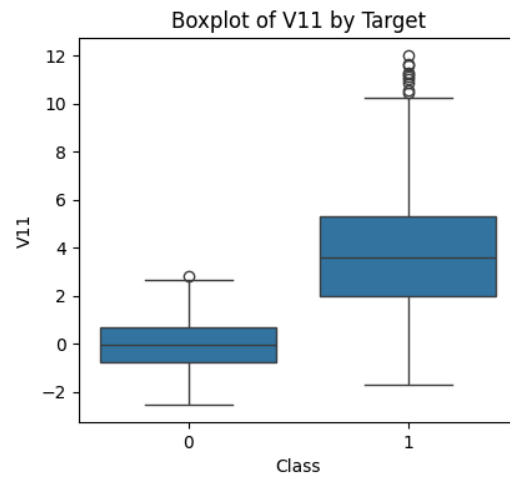
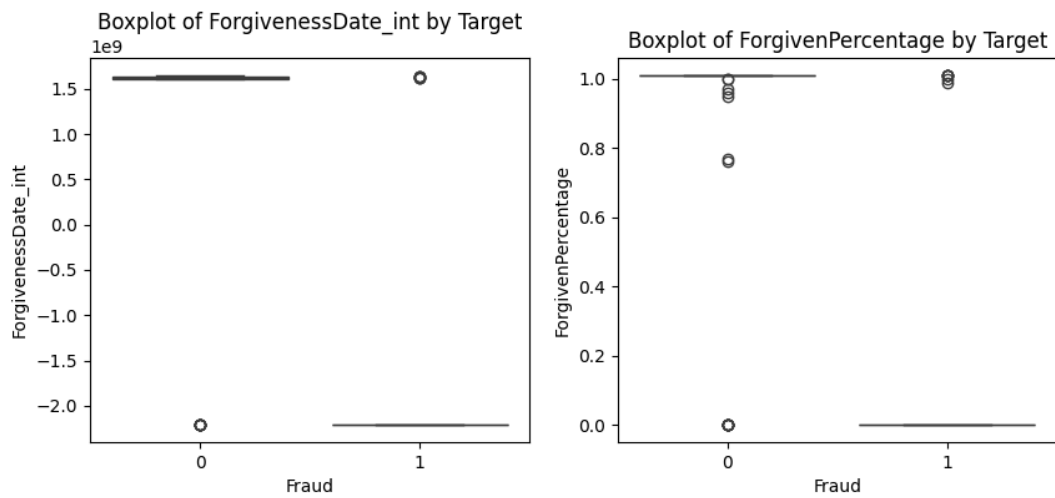


Figure 8: PPP Boxplot of ForgivenessDate and ForgivenessPercentage



References

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AI Appendix

I did not use AI for this submission.